

Part I

SUMMARY of the Fundamental Theorems of Calculus

Summary

February 21, 2026

The First Fundamental Theorem of Calculus:

Given a continuous function f on the interval $[a, b]$, we define an accumulation function, A by

$$A(x) = \int_a^x f(t) dt$$

The First Fundamental Theorem of Calculus (FFTOC) states that

$$A'(x) = f(x)$$

or, equivalently, that

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Note 1: $A(a) = 0$

Note 2: The function A is an antiderivative of f .

Therefore, any antiderivative, G of f can be written as $G(x) = A(x) + C$, or, equivalently,

$$G(x) = \int_a^x f(t) dt + C.$$

Since $G(a) = A(a) + C = 0 + C = C$, it follows that $G(x) = A(x) + G(a)$, or, equivalently, that

$$G(x) = G(a) + \int_a^x f(t) dt$$

Note 3: Let f' , the derivative of f , be continuous on $[a, b]$. Since f is its antiderivative, we have

$$f(x) = f(a) + \int_a^x f'(t) dt$$

The Second Fundamental Theorem of Calculus:

FFTOC implies that, for $x = b$, $A(b) = \int_a^b f(t) dt$. Since $A(a) = 0$, we can write

$$\int_a^b f(t) dt = A(b) - A(a)$$

Note 4: This is the Second Fundamental Theorem of Calculus (SFTOC).

Note 5: Equivalently, using the result in Note 3, for $x = b$, $f(b) = f(a) + \int_a^b f'(t) dt$. So $\int_a^b f'(t) dt = f(b) - f(a)$, or equivalently,

$$\int_a^b \frac{d}{dx} f(x) dx = f(b) - f(a).$$

Part II

Recitation Questions

Problem 1

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Problem 1

Problem 1 *True or False: If f is continuous on the closed interval $[a, b]$, then*

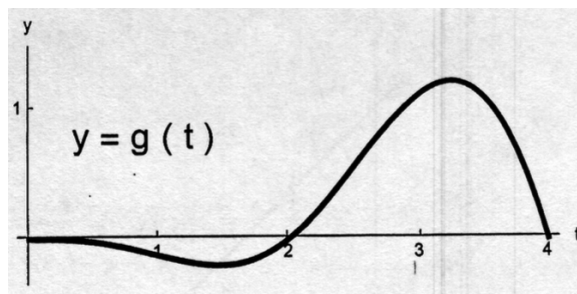
$$\frac{d}{dx} \left(\int_a^b f(t) dt \right) = f(x)$$

Problem 2

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Problem 2

Problem 2 The graph of g , a continuous function on $[0, 4]$, is shown in the figure. Let $A(x) = \int_0^x g(t) dt$, for $0 \leq x \leq 4$.



- (a) Circle the correct statement about $A(2)$.
- (i) $A(2) = 0$;
 - (ii) $A(2) > 0$;
 - (iii) $A(2) < 0$;
 - (iv) none of the previous answers.
- (b) Circle the correct statement about $A(3.8)$.
- (i) $A(3.8) = 0$;
 - (ii) $A(3.8) > 0$;
 - (iii) $A(3.8) < 0$;
 - (iv) none of the previous answers.
- (c) Circle the correct statement about $A'(3.8)$.
- (i) $A'(3.8) = 0$;
 - (ii) $A'(3.8) > 0$;
 - (iii) $A'(3.8) < 0$;
 - (iv) none of the previous answers.
- (d) Find the solution of the following initial value problem: $y'(x) = g(x)$ and $y(0) = 2$.
- (i) $y(x) = g(x)$;
 - (ii) $y(x) = g(x) + 2$;
 - (iii) $y(x) = A(x)$;
 - (iv) $y(x) = A(x) + 2$;
 - (v) $y(x) = g'(x)$;

Problem 2

- (vi) $y(x) = g'(x) + 2$;
- (vii) *none of the previous answers.*

[2]

(e) Find the expression for $\int_0^4 |g(t)| dt$.

- (i) $A(4)$;
- (ii) $A(2) - A(4)$;
- (iii) $A(4) - A(2)$;
- (iv) $A(4) - 2A(2)$;
- (v) *none of the previous answers.*

(f) Find the midpoint Riemann sum for the function g on the interval $[0, 4]$ with $n = 2$ (the number of subintervals).

- (i) $g(1) + g(3)$;
- (ii) $g(0) + g(2)$;
- (iii) $g(2) + g(4)$;
- (iv) $2g(1) + 2g(3)$;
- (v) $2g(0) + 2g(2)$;
- (vi) $2g(2) + 2g(4)$;
- (vii) *none of the previous answers.*

Problem 3

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Problem 3

Problem 3 The limit of Riemann sums for a function f on the interval $[1, 5]$ is given by

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(2x_k^* + \frac{1}{x_k^*} \right) \Delta x \text{ on } [1, 5].$$

- (a) Identify f and express the limit as a definite integral.
- (b) Evaluate the limit of Riemann sums.

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Problem 4

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Problem 4 Compute the following integrals:

(a) $\int_0^1 e^{5x} dx$

(b) $\int_{-2}^{-1} \frac{1}{x^3} dx$

(c) $\int_0^4 \left(3x - 5 + 7\sqrt{16 - x^2} \right) dx$
[3]

Problem 5

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Problem 5 Find the derivative of the following functions:

(a) $F(x) = \int_{\sqrt{x}}^1 \frac{t^2}{2 + 3t^4} dt$

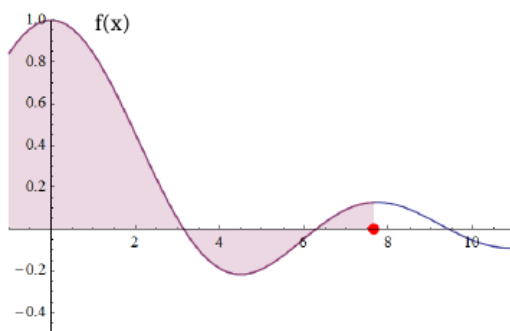
(b) $G(x) = \int_x^{x^3} \sin(7t^2) dt$

Problem 6

February 21, 2026

Problem 6

Problem 6 Given the following graph of $y = f(x)$, let $g(x) = \int_{-1}^x f(t) dt$.



- (a) Is g continuous? Why or why not?
- (b) Is g differentiable? Why or why not?
- (c) Where does g achieve its absolute maximum and minimum values? Where does g achieve any local extreme values? Assume the domain of g is $[-1, 7.6]$.
[4]
- (d) Where is the graph of g concave up? Concave down?
[2]

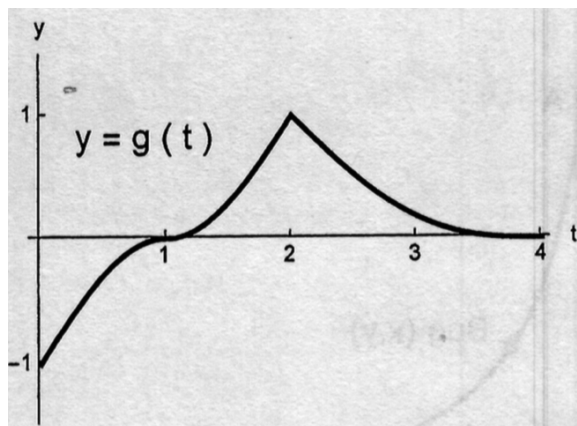
Problem 7

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Problem 7

Problem 7 The graph of g , a continuous function $[0, 4]$, is shown in the figure.

Let $A(x) = \int_0^x g(t) dt$ for $0 \leq x \leq 4$.



(a) Circle the correct statement about $A(1)$.

- (i) $A(1) = 0$;
- (ii) $A(1) < 0$;
- (iii) $A(1) > 0$;
- (iv) none of the previous answers.

(b) Circle the correct statement about $A(1.5)$.

- (i) $A(1.5) = 0$;
- (ii) $A(1.5) < 0$;
- (iii) $A(1.5) > 0$;
- (iv) none of the previous answers.

(c) Circle the correct statement about $A'(1.5)$.

- (i) $A'(1.5) = 0$;
- (ii) $A'(1.5) < 0$;
- (iii) $A'(1.5) > 0$;
- (iv) none of the previous answers.

(d) Circle the correct expression for $\int_1^4 (g(t) + 1) dt$.

- (i) $A(4) + 1$;
- (ii) $A(4) - A(1)$;

Problem 7

- (iii) $A(4) - A(1) + 1$;
- (iv) $A(4) + 3$;
- (v) $A(4) - A(1) + 3$;
- (vi) *none of the previous answers.*

[3]

(e) *Circle the correct statement about $A(0)$.*

- (i) $A(0) = 0$;
- (ii) $A(0) = 1$;
- (iii) $0 < A(0) < 1$;
- (iv) $A(0) > 1$;
- (v) $A(0) < 0$;
- (vi) *none of the previous answers.*

(f) *Circle the interval (or intervals) where the function A is DECREASING.*

- (i) $(0, 1)$;
- (ii) $(1, 2)$;
- (iii) $(2, 4)$;
- (iv) *none of the previous answers.*

(g) *Circle the value (or values) where the function A attains its MAXIMUM.*

- (i) $x = 0$;
- (ii) $x = 1$;
- (iii) $x = 2$;
- (iv) $x = 3$;
- (v) $x = 4$;
- (vi) *none of the previous answers.*

(h) *Circle the value (or values) where the function A attains its MINIMUM.*

- (i) $x = 0$;
- (ii) $x = 1$;
- (iii) $x = 2$;
- (iv) $x = 3$;
- (v) $x = 4$;
- (vi) *none of the previous answers.*

(i) *Circle the interval (or intervals) where the function A is CONCAVE DOWN.*

Problem 7

- (i) $(0, 1)$;
 - (ii) $(1, 2)$;
 - (iii) $(2, 4)$;
 - (iv) *none of the previous answers.*
- (j) *Circle the value (or values) of x where the function A has an inflection point.*
- (i) $x = 1$;
 - (ii) $x = 2$;
 - (iii) $x = 3$;
 - (iv) *none of the previous answers.*
- (k) *Sketch the graph of A , based on items (e)-(j)*